

E-Commerce Marketing Analytics Dashboard - Full Documentation

End-to-End Development Guide: From Architecture & Setup to Complete React Code & Features

1. Project Architecture & Setup

The project is built using React 18, Vite, Tailwind CSS, and Recharts. Below are the core configuration files required for post-processing and utility-first CSS integration.

A. PostCSS Config (*postcss.config.js*)

```
export default {
  plugins: {
    tailwindcss: {},
    autoprefixer: {},
  },
}
```

B. Tailwind Config (*tailwind.config.js*)

```
/** @type {import('tailwindcss').Config} */
export default {
  content: [
    "./index.html",
    "./src/**/*.{js,ts,jsx,tsx}",
  ],
  theme: {
    extend: {},
  },
  plugins: [],
}
```

2. Entry Point & Global Styles

A. Tailwind Directives (*src/index.css*)

```
@tailwind base;
@tailwind components;
@tailwind utilities;
```

B. Application Mounting (*src/main.jsx*)

```
import React from 'react'
import ReactDOM from 'react-dom/client'
import App from './App.jsx'
import './index.css'

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>,
)
```

3. Core Dashboard Application (*src/App.jsx*)

Includes KPI Cards, Dynamic Filtering, Recharts Data Visualizations, Marketing Campaign Response Heatmap, and One-Click CSV Export.

```
import React, { useState } from 'react';
import { BarChart, Bar, XAxis, YAxis, Tooltip, ResponsiveContainer, Cell, PieChart, Pie, Legend } from 'recharts';
```

```

import { DollarSign, ShoppingCart, Users, Target, Filter, Download, Flame } from 'lucide-react';

export default function App() {
  const [selectedCategory, setSelectedCategory] = useState('All');

  const rawCategories = [
    { Category: 'Wines', Total_Spend: 520000, Purchases: 18000, Conversions: 1400, Avg_Income: 72000 },
    { Category: 'Meat', Total_Spend: 380000, Purchases: 12500, Conversions: 1100, Avg_Income: 68000 },
    { Category: 'Gold', Total_Spend: 190000, Purchases: 6200, Conversions: 650, Avg_Income: 75000 },
    { Category: 'Fish', Total_Spend: 140000, Purchases: 4500, Conversions: 420, Avg_Income: 61000 },
    { Category: 'Fruits', Total_Spend: 85000, Purchases: 2500, Conversions: 200, Avg_Income: 55000 },
    { Category: 'Sweets', Total_Spend: 42890, Purchases: 1500, Conversions: 120, Avg_Income: 52000 }
  ];

  // CSV Export Functionality
  const exportToCSV = () => {
    const headers = ["Category,Total_Spend,Purchases,Conversions,Avg_Income\n"];
    const rows = filteredCategories.map(c =>
      `${c.Category},${c.Total_Spend},${c.Purchases},${c.Conversions},${c.Avg_Income}`
    );
    const blob = new Blob([headers.concat(rows.join("\n"))], { type: 'text/csv;charset=utf-8;' });
    const url = URL.createObjectURL(blob);
    const link = document.createElement("a");
    link.setAttribute("href", url);
    link.setAttribute("download", `Marketing_Report_${selectedCategory}.csv`);
    document.body.appendChild(link);
    link.click();
    document.body.removeChild(link);
  };

  // Heatmap Color Scale
  const getHeatmapBg = (val) => {
    if (val >= 80) return 'bg-emerald-500/80 text-white font-bold';
    if (val >= 60) return 'bg-emerald-600/50 text-emerald-100';
    if (val >= 40) return 'bg-amber-500/40 text-amber-100';
    if (val >= 25) return 'bg-slate-700/60 text-slate-300';
    return 'bg-slate-800 text-slate-500';
  };

  return (
    // Component Render UI...
  );
}

```

4. Execution & Server Launch Commands

```

# Navigate to project directory
cd C:\DataAnalystProjects\ecommerce-marketing-analytics-dashboard\05_interactive_web

# Install all Node dependencies
npm install

# Start Vite Development Server
npm run dev

```